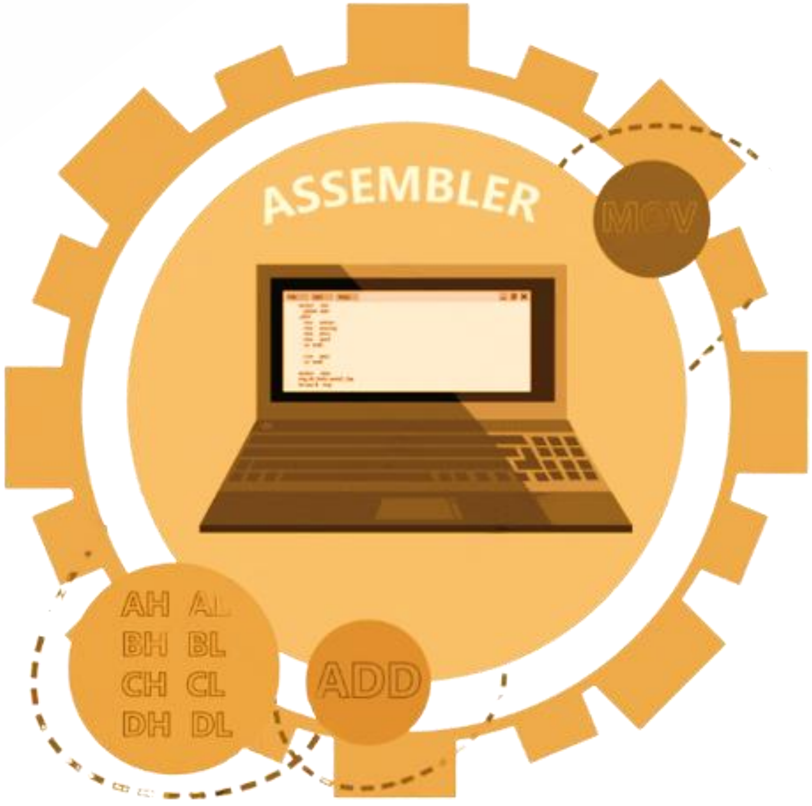
Lecture # 2

Number Systems

in Assembly Language

Engr. Fiaz Khan



Today's Agenda

- ✓Number Systems
- ✓Binary Number System
- ✓Hexadecimal Number System
- ✓1's Complement
- ✓2's Complement
- ✓ASCII Table

Number System

- Binary Number System
- Hexadecimal Number System

Binary Number System

- In the binary number system, the base is two and there are only two digits, 0 and 1.
- For example, the binary string 11010 represents the number

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 26$$

- The base two is represented in binary as 10.

Hexadecimal Number System

- Binary numbers can be long and challenging to express, especially for memory representation.
- 16 bits are required for an 8086-based computer to represent the contents of a memory word.
- Decimal numbers are complex to convert into binary.

Hex Digits and Binary Equivalent

Hex Digits	*Binary*
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Conversion between Number System

Consider the hex number 82AD. It can be written as

$$\begin{aligned} 8A2Dh &= 8 \times 16^3 + A \times 16^2 + 2 \times 16^1 + D \times 16^0 \\ &= 8 \times 16^3 + 10 \times 16^2 + 2 \times 16^1 + 13 \times 16^0 = 35373d \end{aligned}$$

Similarly, the binary number 11101 may be written as

$$11101b = 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 29d$$

This gives one way to convert a binary or hex number to decimal, but an easier way is to use nested multiplication. For example,

$$\begin{aligned} 8A2D &= 8 \times 16^3 + A \times 16^2 + 2 \times 16^1 + D \times 16^0 \\ &= ((8 \times 16 + A) \times 16 + 2) \times 16 + D \\ &= ((8 \times 16 + 10) \times 16 + 2) \times 16 + 13 \\ &= 35373d \end{aligned}$$

Example:

- Convert 11101 to decimal.

Example 2:

- Convert 2BD4h to decimal.

The one's complement

- The one's complement of an integer is obtained by complementing each bit; that is, replace each 0 by a 1 and each 1 by a O.

Find the one's complement of 5 = 0000000000000101.

5 = 0000000000000101
One's complement of 5 = 1111111111111010

Two's Complement

- To get the two's complement of an integer, just add 1 to its one's complement.

$$
\begin{array}{rr}
\text{one's complement of } 5 = & 1111111111111010 \\
& +\ 1 \\
\hline
\text{two's complement of } 5 = & 1111111111111011
\end{array}
$$

Simple way of finding Two's Complement

- From right to left, leave all zeros and first one, invert the rest bits.
- 10010100
- 01101100

ASCII (American Standard Code for Information Interchange)

- ASCII (American Standard Code for Information Interchange) is a popular encoding scheme for characters used by computers.
- It originally originated in teletype communications but is now standard for personal computers.
- ASCII codes use seven bits for each character, providing a total of 128 codes (2^7).
- Table 2.5 lists ASCII codes and associated characters, with only 95 codes (32 to 126) considered printable.
- Non-printable codes (0 to 31 and 127) were initially for communication control purposes.

ASCII (American Standard Code for Information Interchange)

- Non-printable codes (0 to 31 and 127) were initially for communication control purposes.
- Most microcomputers use only printable characters and a few control characters like LF, CR, BS, and Bell.
- Each ASCII character is encoded with seven bits, fitting into a byte with the most significant bit set to zero.
- Printable characters can be displayed on monitors or printed, while control characters control device operations.
- Special display characters can be assigned to non-printed ASCII codes, allowing extended character sets on some computers.
- The screen controller for the IBM PC, for example, can display an extended set of 256 characters, as shown in Appendix A.

ASCII Table

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

Source: www.LookupTables.com

THANK YOU!

Do you have any questions?

ENGR. FIAZ KHAN

LECTURER COMPUTER SCIENCE

TELF/EF SET / ACEPT CERTIFIED

MICROSOFTRAINER

ORACLE CERTIFIED PROFESSIONAL

T ACADEMY GOOGLE CERTIFIED PROFESSIONAL

Lahore, Pakistan

+92-320-7617-093

Muhammad.fiaz@cs.uol.edu.pk

https://www.linkedin/in/fiazofficials